

Ultrasonic Cutting of Nonwoven Mats

Findings and equipment recommendations — Fybron, FiberCom, Fiberite

Rev. B — what changed

Three peer-reviewed sources [18][19][20] confirmed the two central claims of rev. A and **corrected three parameter recommendations**: traverse speed was too fast by 3–5x, “blade angle” conflated two distinct angles, and horn material was over-specified to titanium. Two new first-order findings were added — wide-blade **amplitude uniformity** and **through-thickness energy transfer**. Corrected rows are marked in red.

Bottom line

- The nominal **50% thermoplastic** content overstates what melts at the kerf. Effective sealing content is **~50% for Fiberite**, only **~15% for Fybron and FiberCom**. Two published thresholds bracket this: ~50% to cut and seal [1], 60–65% to weld [18]. [1][18][17]
- A **split melt range is itself a named limitation**. Kara lists exactly three barriers to ultrasonic weldability — high unit mass, high thickness, and “fiber content with two different melting temperatures.” That study’s fibres melt ~30 °C apart; your CoPET/nylon split is **145 °C**. All three barriers apply to your material at once. [18]
- These are not textile nonwovens — now quantified**. The paper titled “thick and heavier nonwoven fabrics” tops out at **227.9 gsm / 0.83 mm**. That is lighter than your *lightest* material. Fiberite is 8x the basis weight and 3.5x the thickness at a third of the meltable content. [18][17]
- The **U.S. Solid 28 kHz / 500 W unit on site** still answers go/no-go this week, but at 500 W it is **genuinely marginal for Fiberite** — Kim needed 1,200 W for a comparable-difficulty cut. [19]

1 · Only part of the “thermoplastic” fraction melts

Constituent	Fybron	FiberCom	Fiberite	Behaviour at the kerf	Seals
CoPET binder (110 °C)	15%	15%	35%	flows well below cellulose char	yes
PLA (Kilop 6 mm)	—	—	15%	melts ~150–170 °C	yes
Nylon 6,6	35%	35%	—	~255 °C — above char onset	marginal
Lyocell / viscose / pulp / hemp	50%	50%	~33%	never melts; chars 200–300 °C	no
Carbon fibre (Vartega)	—	—	15%	inert, hard, abrasive	no
Effective sealing fraction	15%	15%	50%		

The press schedule is the evidence: Fybron consolidates at 160 °C and Fiberite at 180 °C [17], both far below nylon’s melt. Nylon is a reinforcing fibre here and will not become a binder under an ultrasonic blade — the cellulose beside it chars first. The thickest, densest, carbon-loaded material is the best ultrasonic candidate.

2 · Wrong regime for published data

	Spunbond (typical study)	Kara “thick and heavier” [18]	Simplifyber mats [17]
Basis weight	20–100 gsm	108.9–227.9 gsm	350–1,900 gsm
Thickness	0.1–0.5 mm	0.39–0.83 mm	0.65–2.9 mm
Density	50–200 kg/m³	—	430–910 kg/m³
Thermoplastic	100%	100% (PET:PA6 bico)	15–50% low-melt

2a · New: energy may not reach the far side

Kara’s two heaviest fabrics — again, only 0.69 and 0.83 mm — produced seams whose “face and back sides were quite different, that proved the inefficient energy transfer of the welding cylinder for these samples.” Ultrasonic energy was already failing to

penetrate sub-millimetre material at 34.9 kHz. At 0.65–2.9 mm and half the meltable content, expect this to be the dominant failure mode rather than an edge case. [18]

- **Use 20 kHz, not 28 or 35 kHz.** Lower frequency, higher amplitude, deeper penetration. This is now the best-supported parameter choice in the brief.
- **Expect asymmetric edge quality** — top face sealed, bottom fibrous. Inspect *both* faces of every test cut; judging from one side will mislead.
- **Power must scale with thickness.** 125–200 W sufficed for 0.4–0.8 mm of fully thermoplastic fabric [18]; 1,200 W was needed for a 6.82 mm carbon preform [19]. Your material sits between them. [18][19]

3 · Parameters to design to

Parameter	Value	Source / rationale
Frequency	20 kHz for Fyberite and all coated stock; 28–40 kHz only for thin Fybron	Higher amplitude and deeper penetration; directly supports §2a. Patent art spans 15–100 kHz. [1][6][18]
Amplitude	30 µm wide-blade nonwoven work; 60 µm carbon-loaded Fyberite	Lee's 20 kHz wide-blade horn delivers 22–30.6 µm [20]; Kim cut carbon preform at 60 µm [19].
Amplitude (relative)	70–100% of rated; default 100%	Below 70% not recommended without consultation. [8]
Power	500 W screening only; 1–2 kW production on Fyberite	125–200 W for 0.8 mm fully thermoplastic [18]; 1,200 W for 6.82 mm carbon [19].
Traverse speed	Start at 1 m/min. Increase only while edge quality holds	Corrected from rev. A (was 3–5 m/min). Kim found 1 m/min gave zero damage; at 3 and 5 m/min the knife “could not cut the preform plate effectively” and tows were dislodged [19]. Kara welded at 1.7–2.0 m/min [18].
Blade point angle	24° cemented carbide for Fyberite; 30–55° steel for Fybron	Rev. A conflated two angles. This is the included angle of the edge itself. Kim used a 24° carbide cutter on carbon [19]; Eastman drag blades are 30/45/55° [10].
Knife attack angle	45° — the shallowest tested	Distinct from point angle: the angle between knife and workpiece. Feed force rises and thrust force falls as attack angle increases; 75° was worst. [19]
Horn material	Tool steel for a bulky wide-blade horn; Ti-6Al-4V for slender high-cycle horns	Corrected from rev. A (was titanium throughout). Lee specifies tool steel (E 242.5 GPa, HRC 68) for stiffness and stability in a wide blade. [20]
Anvil geometry	Sealing surface 3–20° included, preferably 5–15° ; reinforcing surface $\geq 2.5^\circ$ larger, $\leq 45^\circ$, preferably $\sim 30^\circ$	Shallower sealing angle widens the seal band. [2]
Blade–anvil gap	~0.003 in (76 µm), set then locked	Gap consistency dominates edge quality. [3]
Backing (free cut)	Silicone rubber sacrificial plate; over-travel past sheet thickness	Kim backed a 6.82 mm preform with silicone rubber, cutting to 9 mm depth. [19]
Ramp-up time	0.150 s @ 20 kHz · 0.100 s @ 30 kHz · 0.050 s @ 40 kHz	The blade must not contact material during ramp-up. [8][9]
Expected tolerance	+250 to +365 µm oversize, best case	Kim's best condition on carbon preform; treat as the floor, not the target. [19]

3a · New: a wide blade does not vibrate evenly

For panel blanking the natural tool is a **wide-blade horn**, which acts as the punch in a punching and blanking system. Lee designed, tuned and measured one — and the result is a warning. [20]

Property	Measured
Configuration	20 kHz, 180 mm wide, tool steel; converter 18 µm in → 30.6 µm out (1 : 1.5 gain)
Amplitude uniformity	76% only — 22.0 µm minimum against 29.1 µm maximum along the edge
Distribution	M-shaped : maxima about 2/3 out from each end, minimum in the middle
Slots	8 mm wide x 79 mm long, 62 mm apart — improve uniformity <i>and</i> radiate heat, preventing hot spots
Tuning reality	fabricated horn landed 492 Hz off FEA; heat treatment dropped it 300 Hz; polishing the output face raised it 29 Hz

A 24% amplitude deficit mid-blade matters enormously for material sitting at the sealing threshold. On Fyberite the low spots may still seal; on Fybron, at ~15% low-melt, the difference between 29 µm and 22 µm is plausibly the difference between a sealed edge and a furry one. **Make amplitude uniformity a written spec** — ask any wide-blade vendor for the measured minimum-to-maximum ratio along the edge, and for slot geometry. Do not accept a nominal amplitude figure alone.

4 · Two configurations, because there are two jobs

	Flat panel blanking	Trimming formed 3D parts
Support	blade against a hard anvil	free cutting onto compliant sacrificial backing
Edge result	cut and sealed — the anvil concentrates energy	severing only; no seal, nothing to react against
Key control	anvil angle and gap [2][3]; blade amplitude uniformity §3a [20]	blade attitude tangent to path — 7th axis — plus radial compliance to follow the flash line [4][15]; narrow parameter window [19]
Buy when	now, after screening	only after Tier 1 data

5 · Equipment ladder

Tier	What	Notes
0 — own it	U.S. Solid 28 kHz / 500 W handheld (~\$995)	Answers go/no-go this week at zero further cost, but 500 W is marginal for Fyberite per §2a. Handheld, fixed amplitude, compromise frequency — screening only. Add a rigid fixture, <i>both</i> a hard anvil and a silicone rubber backing plate, a guide fence, fume extraction and hearing protection. [19]
1 — buy next	Ultrasonic tool head on a CNC flatbed (<i>recommended</i>)	Eastman: 20 kHz spindle, 30/45/55° drag blades, static-table and conveyor systems. Controlled path, depth and speed in one purchase; the same table takes mechanical tools so ultrasonic can be benchmarked on identical geometry. Zünd lists no ultrasonic module — verify before comparing quotes. Specify amplitude uniformity in the PO. [10][11][20]
1 — alt.	Component stack, self-integrated	20 kHz generator (1–2 kW) + converter + booster + custom blade horn from Dukane, Sonics & Materials, Branson/Emerson, Telsonic, Herrmann, Rinco or Sinaptec. Cheaper in parts, but horn design is the hard part — Lee needed FEA plus two physical tuning iterations to land within 21 Hz of target. Not a purchasing exercise. [8][12][20]
2 — later	Robotic 3D trim cell	6-axis robot, compliant wrist, 7th axis for blade attitude. Yaskawa Motoman, Shape Process Automation, 4D Systems, PushCorp. Kim's rig — Stäubli RX160 + Sonotec HP-8701 (1,200 W, 24 kHz, 60 µm) + 400 N force sensor — is a validated reference configuration. [4][14][15][19]

6 · Risks

Risk	Why	Mitigation
Through-thickness energy transfer	inefficient already at 0.83 mm [18]; your material is 1–3.5x thicker with half the meltable content	20 kHz, 1–2 kW; inspect both faces; accept asymmetry or cut from both sides
Amplitude non-uniformity	76% min/max along a 180 mm wide blade [20]	specify measured uniformity and slot geometry; validate seal along the whole cut, not at one point
Carbon abrasion	15% Vartega carbon in Fyberite; CFRP is known for short tool life	cemented carbide, 24° point [19]; budget blades as a consumable; track cuts-to-degradation
Cellulose char	33–50% of every formulation cannot melt, only scorch	keep traverse up, amplitude high, dwell low — a brown edge means too slow
Weak seal	~15% low-melt in Fybron and FyberCom — below both published thresholds	do not promise a sealed edge; consider a CoPET-rich scrim or veil at cut lines
Loss of extensibility	strain fell from 66% to ~10–12% across ultrasonic seams [18]	measure elongation, not just strength — a stiff edge may fail a wear test that tensile passes
Fumes	PLA, CoPET, PU coating and pigment decompose at the kerf	local extraction at the cut line; review coating SDS
Noise	20 kHz sits at the edge of hearing and is loud in air	enclose the cell; OSHA 85 dBA 8-hour TWA as the hearing-conservation trigger
Vendor data mismatch	published work stops at 228 gsm / 0.83 mm [18]	send your own coated samples; refuse to buy on a demo of someone else's material

7 · Recommended sequence

- This week, zero cost.** Screen all three materials on the U.S. Solid cutter — coated and uncoated, hard anvil and silicone backing, **starting at 1 m/min**. Measure tensile, tear *and elongation* on a cut edge against a die-cut control, and inspect **both faces** for seal. Results drop into the existing testing workbook. [17][18][19]
- Then send samples out.** Eastman plus one stack vendor. Require they cut *coated* stock at 20 kHz and report on both cut faces.
- Then buy Tier 1**, weighted to whichever product dominates volume, with amplitude uniformity written into the purchase order. [20]
- Keep a mechanical control throughout.** If ultrasonic only severs Fybron without sealing it, a steel-rule die or oscillating knife may be cheaper and faster for that product. Ultrasonic earns its price on Fyberite — the sealed edge, the abrasive-material tool life and the 3D trim path all point there.

Citations

Sources [1]–[9] and [18]–[20] are held in full text in the RAGFlow dataset “**Ultrasonic Cutting**”.

[18] Kara, S. (2024) — *Ultrasonic weldability of thick and heavier nonwoven fabrics*. J. Appl. Polym. Sci. 141(33):e55840. doi:10.1002/app.55840. Five PET:PA6 bicomponent hydroentangled nonwovens, 108.9–227.9 gsm / 0.39–0.83 mm; 34,917 Hz; 125–200 W, 100–200 N, 1.7–2.0 m/min; 60 sample types after ~110 pre-trials. Source of the 60–65% welding threshold, the three stated weldability limitations, the inefficient-energy-transfer finding and the strain-loss data.

[19] Kim, H.G., Hong, T.H., Kim, D., Kim, S.H. (2024) — *An experimental study of ultrasonic-knife cutting for a woven carbon fiber preform by an industrial robot*. Manufacturing Letters 41:581–587 (NAMRC 52). Stäubli RX160 + Sonotec HP-8701, 1,200 W / 24 kHz / 60 µm, 24° cemented carbide, silicone rubber backing; feed 1–5 m/min x attack angle 45–75°. Source of the corrected traverse speed, the attack-angle finding and the dimensional-tolerance figure.

[20] Lee, C.H., Seo, J.S., Park, D.S. (2014) — *Design and Tuning of the Ultrasonic Wide-blade Horn for Joining and Cutting Non-woven Fabrics*. Advanced Materials Research 941–944:1932–1936. doi:10.4028/www.scientific.net/AMR.941-944.1932. 20 kHz tool-steel wide-blade horn, 180 mm, final resonance 19,979 Hz, amplitude uniformity 76%. Source of the amplitude-uniformity finding and the corrected horn material.

[1] US 4,560,427 — *Ultrasonic seal and cut method and apparatus*, 1985. “Typically 50 percent or more” thermoplastic fibre; range 16–100 kHz.

[2] US 4,542,771 — *Adjustable anvil for ultrasonic material cutting and sealing apparatus*, 1985. Anvil sealing surface 3–20° included (preferably 5–15°); reinforcing surface at least 2½° larger, no greater than 45°.

[3] US 5,265,508 — *Ultrasonic cutting system for stock material*, 1993. Shoe-to-anvil clearance approximately 0.003 in; 40 kHz stack.

[4] US 8,277,282 B2 — *Ultrasonic trimming apparatus and ultrasonic trimming method*, 2012. Seventh axis added to a six-axis robot so the blade stays aligned along the cutting direction.

[5] US 5,061,331 — *Ultrasonic cutting and edge sealing of thermoplastic material*, 1991.

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- [6] US 5,948,208 — *Ultrasonic device for cutting a fusible textile web and simultaneously heat-sealing the cut edges*, 1999. Transducer 15–100 kHz.
 - [7] US 10,654,187 B2 — *Device and method for the ultrasonic cutting of workpieces*, 2020. Cutting edge driven at 20–40 kHz.
 - [8] Dukane Application Note AN503 — *Cutting Food Products or Materials with Ultrasonic Blade Horns*, rev. 02, 2016. Amplitude 70–100%; ramp-up 0.150 / 0.100 / 0.050 s at 20 / 30 / 40 kHz; cutting must begin after ramp-up completes.
 - [9] Dukane Application Note AN512 — *Stack Scan*. Locating the free-running frequency to avoid frequency- and peak-overload alarms.
 - [10] Eastman Machine Company — Ultrasonic Tool Head. 20,000 Hz spindle; 30°, 45° and 55° drag blades; static-table and conveyor systems. eastmancuts.com/products/ultrasonic-tool-head/
 - [11] Zünd — Modules and Tools. Composite cutting served by PRT, POT, EOT-250, WKT and ELOSTO mechanical tools; **no ultrasonic module listed**. zund.com/en/cutting-systems/modules-and-tools
 - [12] TELSONIC Ultrasonics — Cutting and Cut'n'Seal with ultrasonics. telsonic.com/en/cutting-with-ultrasonics/
 - [13] Sonobond Ultrasonics — Nonwovens and textiles bonding FAQ: the higher the synthetic content, the easier a material is to cut and seal ultrasonically.
 - [14] Yaskawa Motoman — Robotic ultrasonic cutting systems for plastics, composites and nonwovens. motoman.com/en-us/products/systems/ultrasonic-cutting
 - [15] Shape Process Automation — Ultrasonic knife trimming and deflashing; radial compliant knife follows the flash line despite part shrinkage.
 - [16] Sinosonics — Ultrasonic cutting: frequencies, blades and materials. Bands 20 / 25 / 30 / 35–40 kHz; 100–500 W laboratory, 800 W–3 kW production.
 - [17] Internal — *Material Samples_Formulations & Testing.xlsx*, July 2026 runs. Formulation and testing tabs supplying gsm, thickness, density, tensile and tear; press schedules 160 °C (Fybron) and 180 °C (Fyberite / FyberCom).

Still outstanding: A study of the ultrasonic-vibration cutting of carbon-fibre reinforced plastics (J. Mater. Process. Technol. 1994); the U.S. Solid unit's manual (amplitude and blade specs are unpublished); the Witcobond coating SDS.